

System Dynamics Modeling in AnyLogic

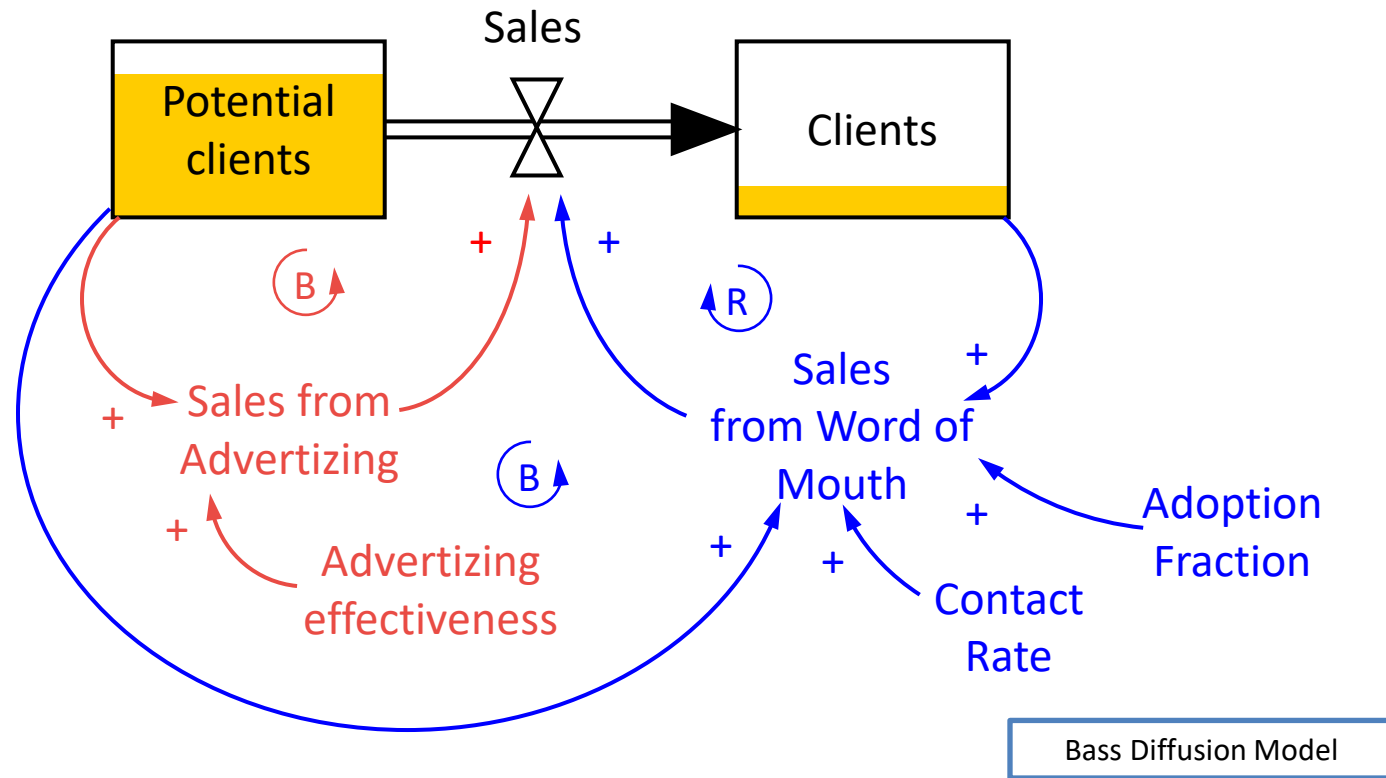
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System Dynamics. Jay Forrester '50s

- Stocks, flows
 - Interacting feedback loops



System Dynamics. Jay Forrester '50s

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 - Interacting feedback loops

The equivalent mathematical model:

$$d(\text{Potential clients})/dt = - \text{Sales}$$

$$d(\text{Clients})/dt = \text{Sales}$$

$$\text{Sales} = \text{Sales from Advertizing} + \text{Sales from Word of Mouth}$$

$$\text{Sales from Advertizing} = \text{Potential clients} * \text{Advertizing effectiveness}$$

$$\begin{aligned} \text{Sales from Word of Mouth} = \\ & \text{Clients} * \text{Contact Rate} * \\ & \left(\text{Potential clients} / (\text{Potential clients} + \text{Clients}) \right) * \text{Adoption Fraction} \end{aligned}$$

Bass Diffusion Model

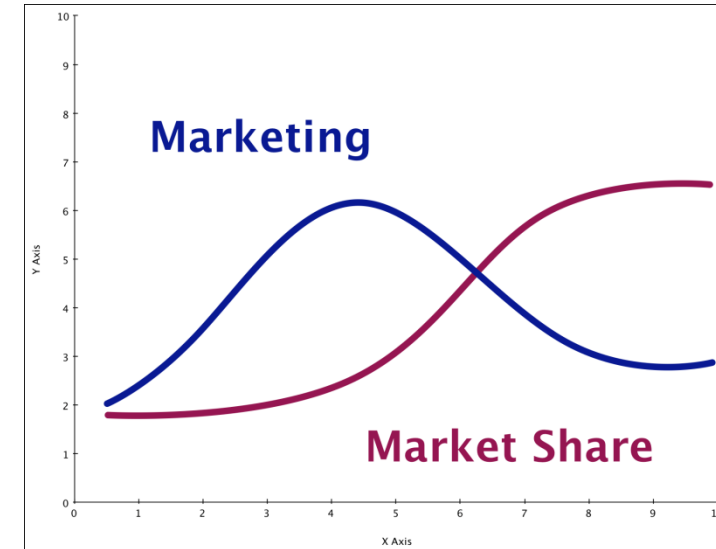
System Dynamics Excels at Certain Things...

- Cause and Effect
 - Representing Causality and relationships
 - System dynamics allow you to represent non-physical relationships too
 - *Morale -> Productivity, Advertising -> Perception,*



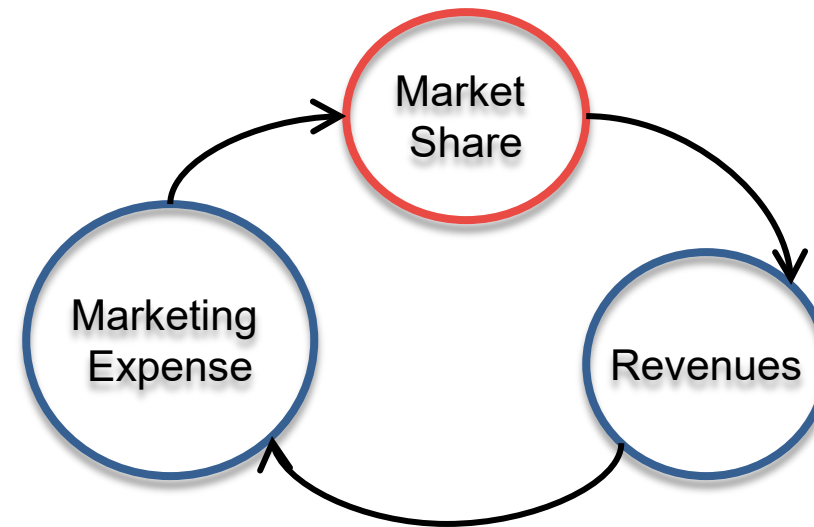
System Dynamics Excels at Certain Things...

- Delays
 - It takes time for certain effects
 - This can be from the nature of the elasticity between variables or from the effects of other mitigating effects
 - *Advertising -> Perception, Actions -> Reputation, Fame -> Perception*

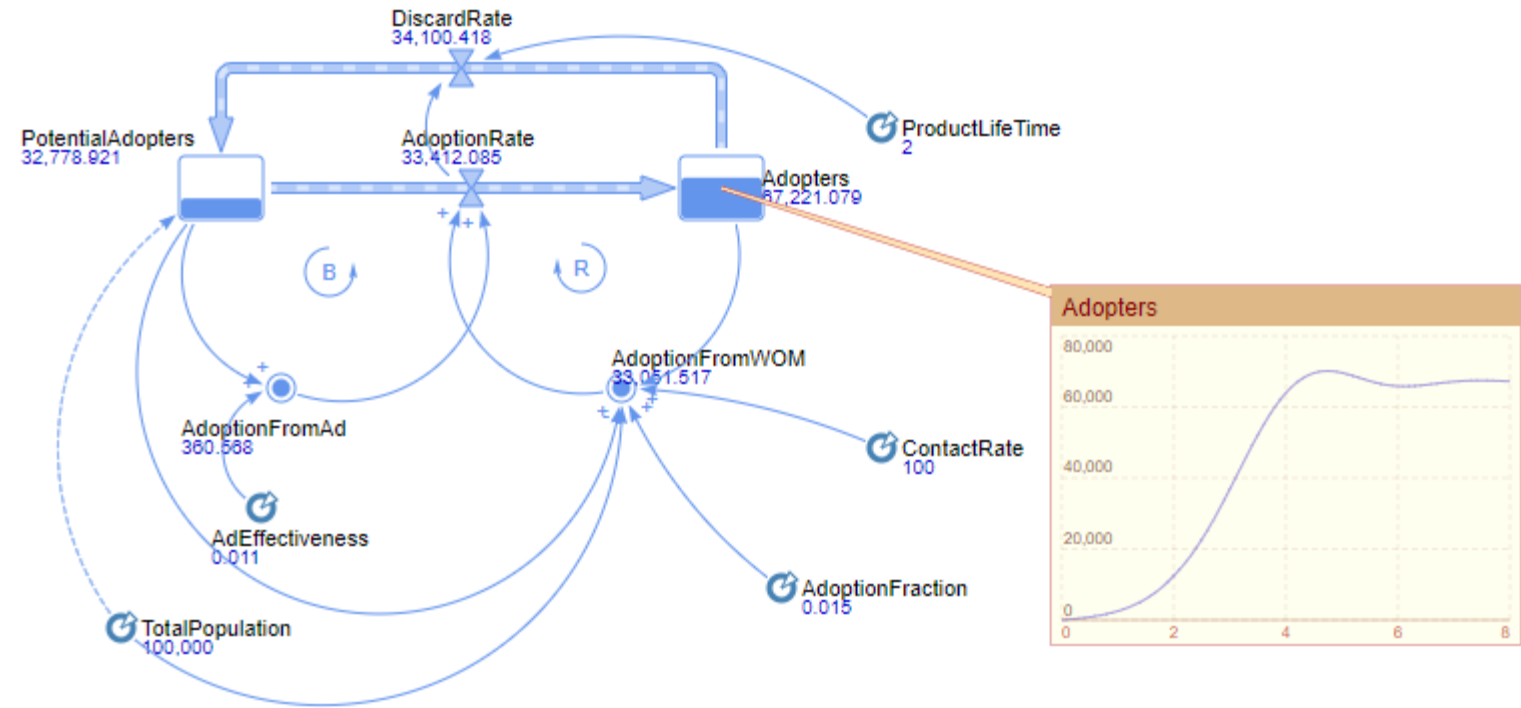
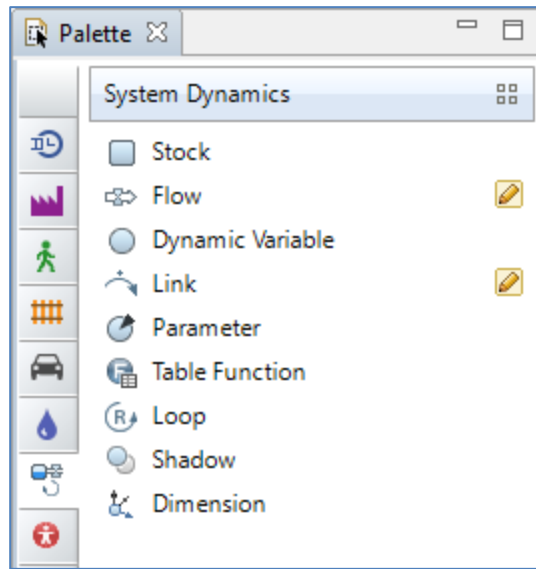


System Dynamics Excels at Certain Things...

- Feedback Effects
 - “Practically” unique to System Dynamics, but common in the real world
 - A logic error in excel
 - Reinforcing Feedback Loops and Correcting Feedback

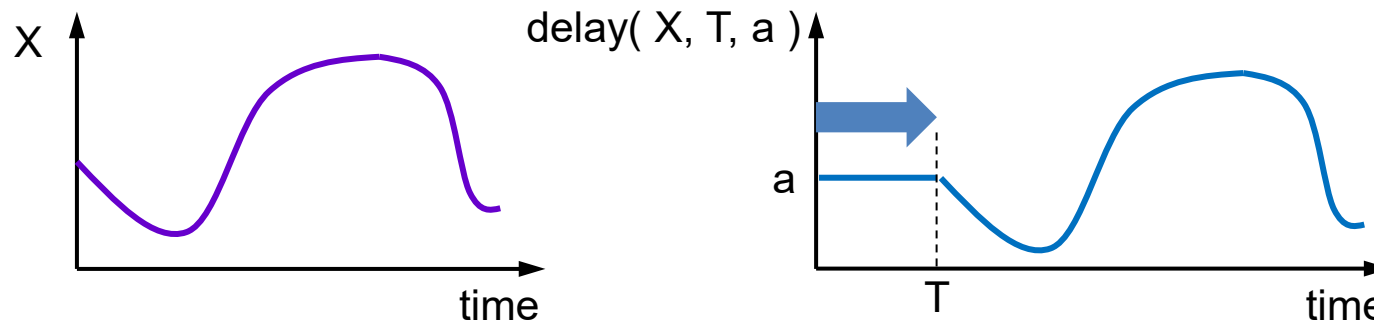


Stock & Flow Elements in AnyLogic



Built-in Functions You Can Use

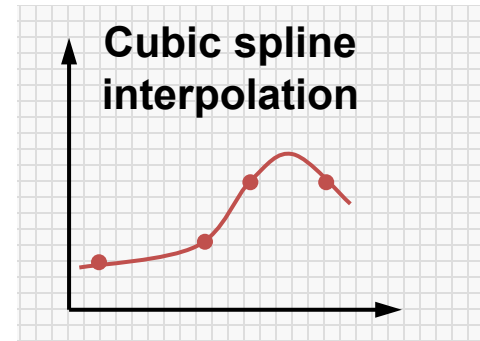
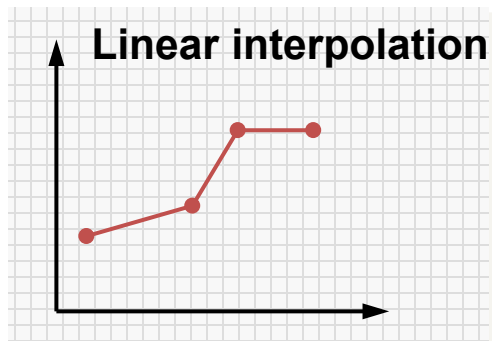
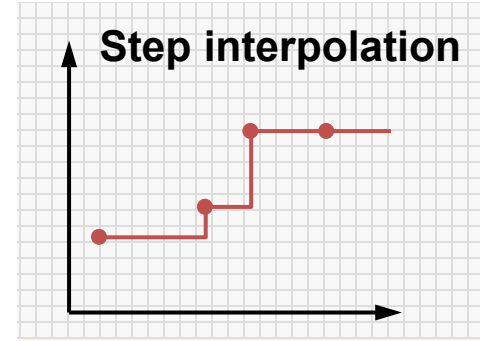
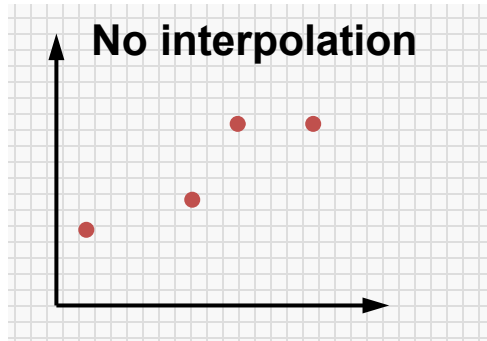
- Mathematical functions
 - `abs(x)` `cos(x)` `exp(x)` `floor(x)` `limit(min,x,max)` `log(x)` `max(a,b)` `min(a,b)` `pow(x)` `round(x)` `sin(x)` `sqrt(x)` `tan(x)` ...
- Special system dynamics functions
 - `delay()` `delay1()` `delay3()` `delayInformation()` `delayMaterial()` `forecast()` `npv()` `npve()` `smooth()` `ramp()` `smooth3()` `trend()` ...



- See full list of functions in AnyLogic Help -> Advanced Modeling with Java -> AnyLogic functions -> System dynamics functions

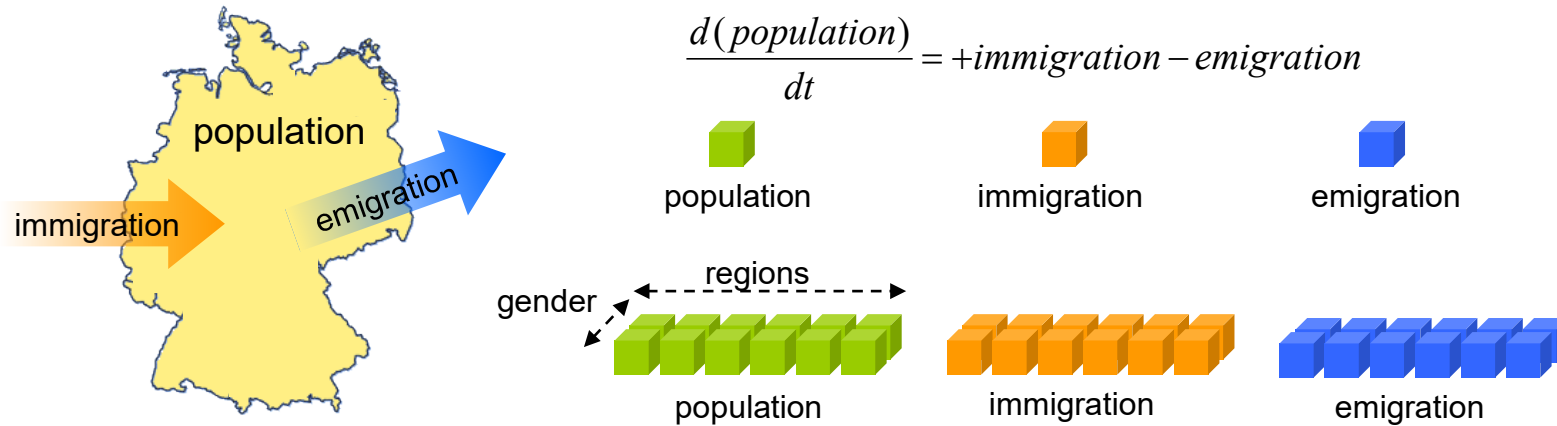
Table Function (Lookup Table)

- Allows to import simple two-column data table into the model and use it as a function:
`myTable(x)`
- Supports various interpolations and approximation



Arrays (“Subscripts”)

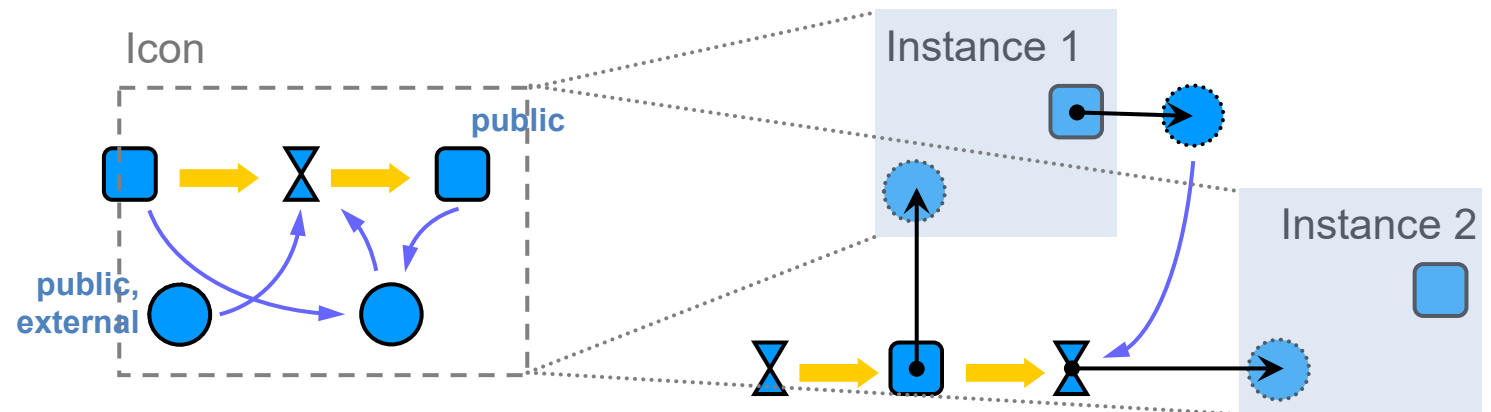
- Some problems require multidimensional data



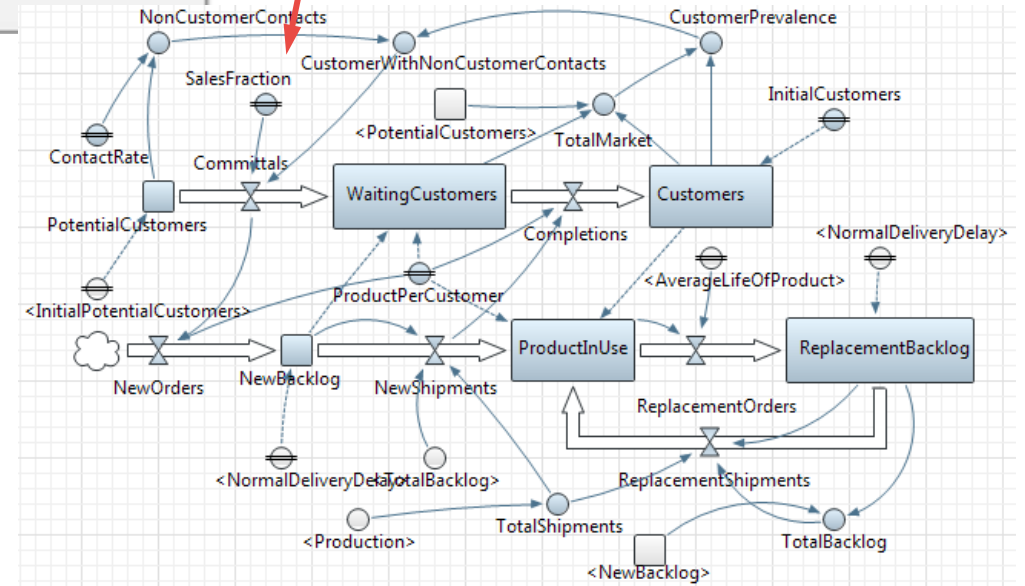
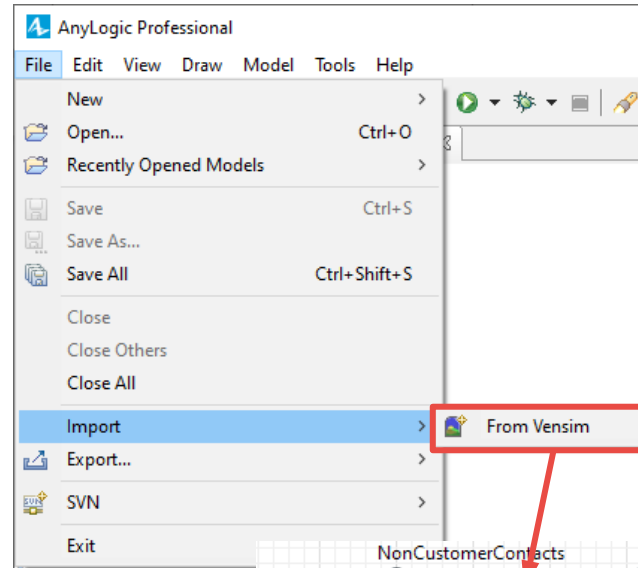
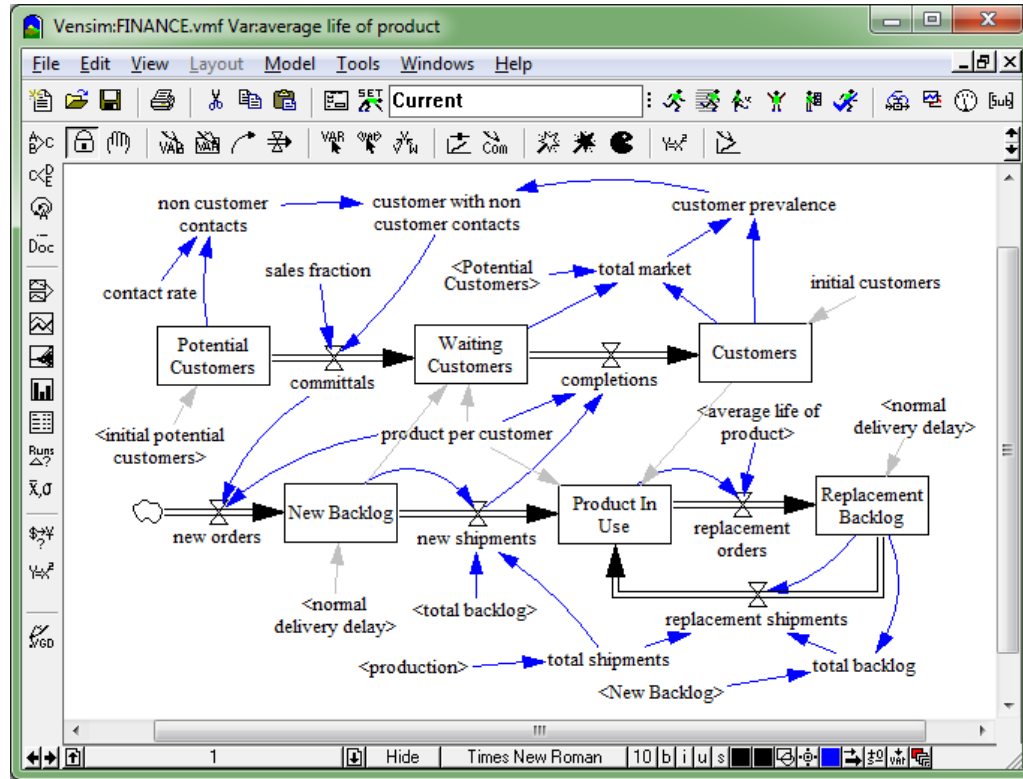
- AnyLogic supports array variables and equations
 - May have any number of dimensions
 - All arithmetic operations and most functions are supported
 - All operations are element-by-element, unlike in linear algebra

Object-Oriented SD Modeling

- Originally classical SD tools:
 - Only allowed you to build one large flat stock & flow diagram which easily could become unmanageable
- AnyLogic:
 - Naturally suggests you to build well-structured models
 - You can build hierarchical models
 - You can create reusable components of SD models

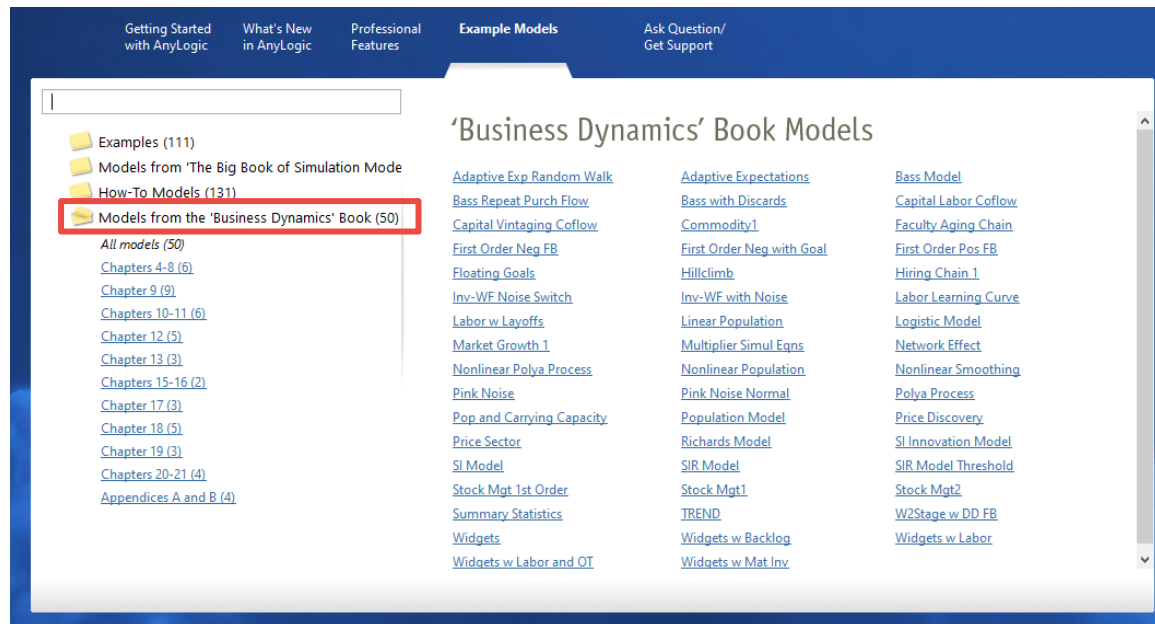
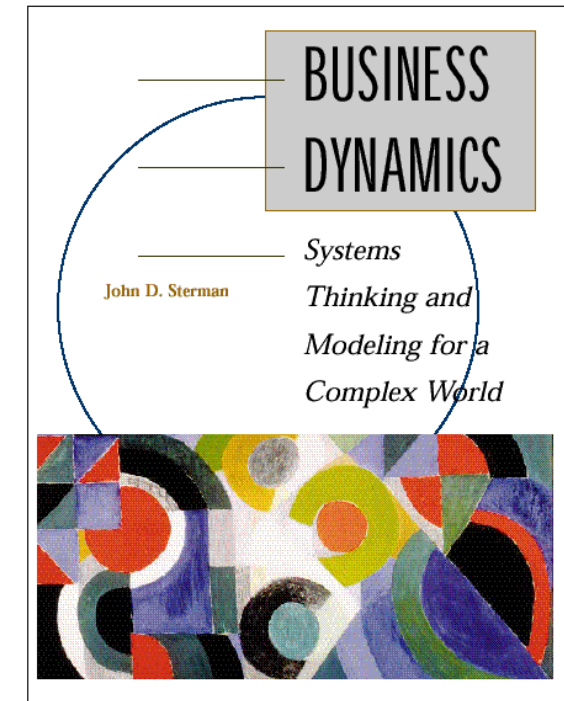


Import of Vensim® Models



The Sacred Book of System Dynamics Modeler

- John D. Sterman
“Business Dynamics:
Systems Thinking and Modeling for a Complex World”
2000. McGraw Hill, 1008 pages
- <http://www.anylogic.com/business-dynamics-book-models>



Bass Diffusion Model

This presentation is a part of AnyLogic Standard Training Program



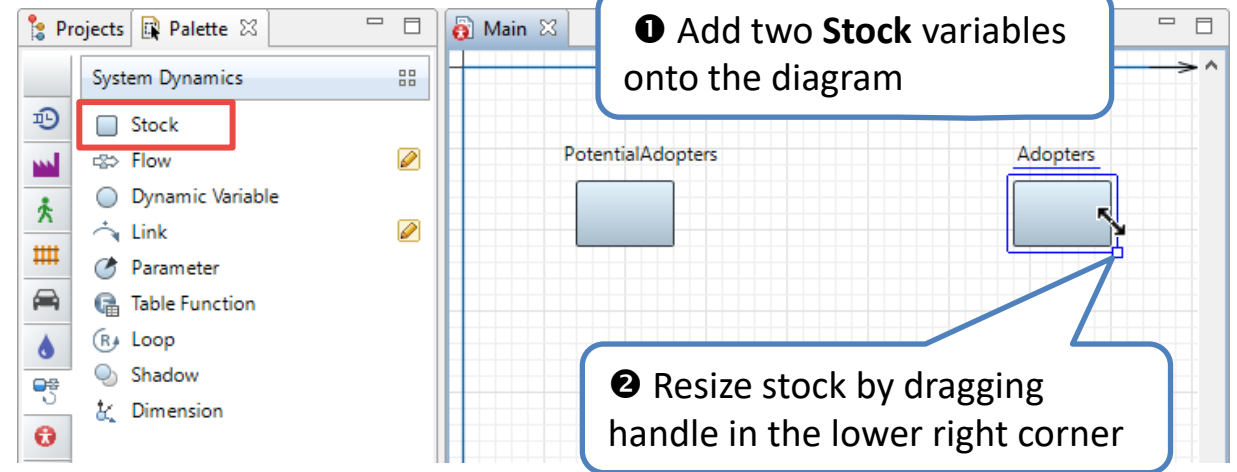
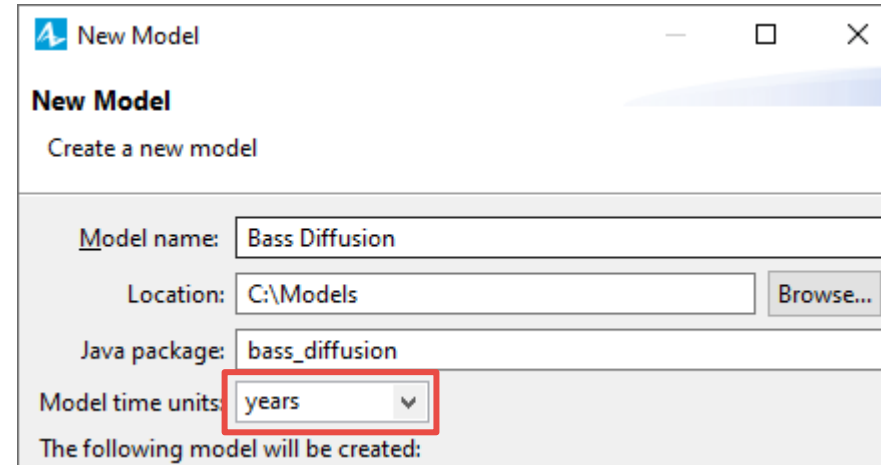
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Bass Diffusion. Phase 1

- SD classical textbook model of product, or innovation diffusion
- A population group is considered to consist of **Potential Adopters** and **Adopters**; all people behave exactly same way
- **Potential Adopters** become **Adopters** at **Adoption Rate** which depends on advertising
- Advertising goes on all the time, and every time unit it converts **Advertising Effectiveness** part of **Potential Adopters** into **Adopters**
- Initially:
 - *Potential Adopters = 100 000*
 - *Adopters = 0*
- Parameter values:
 - *Advertising Effectiveness = 0.011*

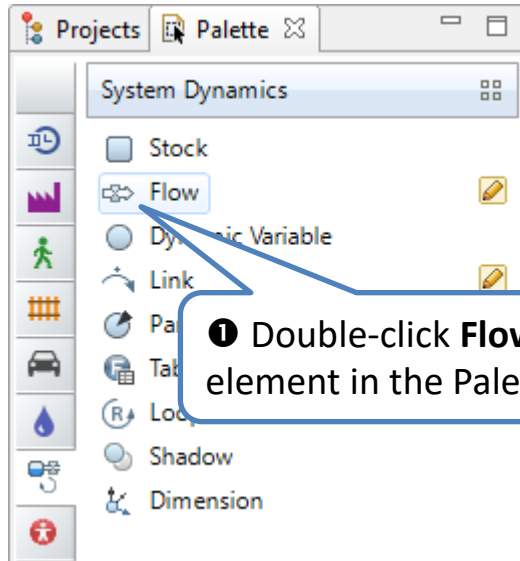
Bass Diffusion. Phase 1. Step 1

- Create new model and name it *Bass Diffusion*
- Set *years* as the model time units
- Open the **System Dynamics** palette
- Add two Stocks onto the diagram
- Name them *PotentialAdopters* and *Adopters*

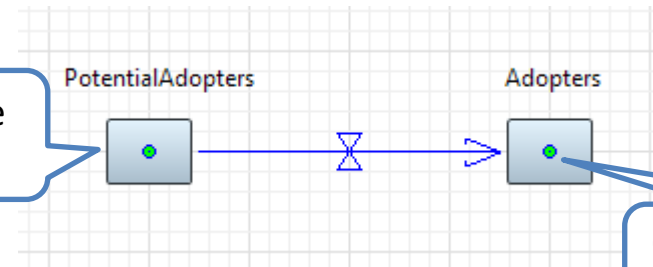


Bass Diffusion. Phase 1. Step 2

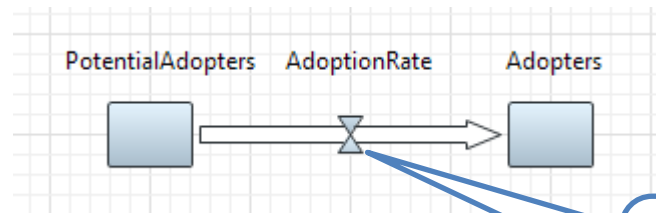
- Create adoption flow from **PotentialAdopters** to **Adopters**



2 Click the stock where the flow flows out



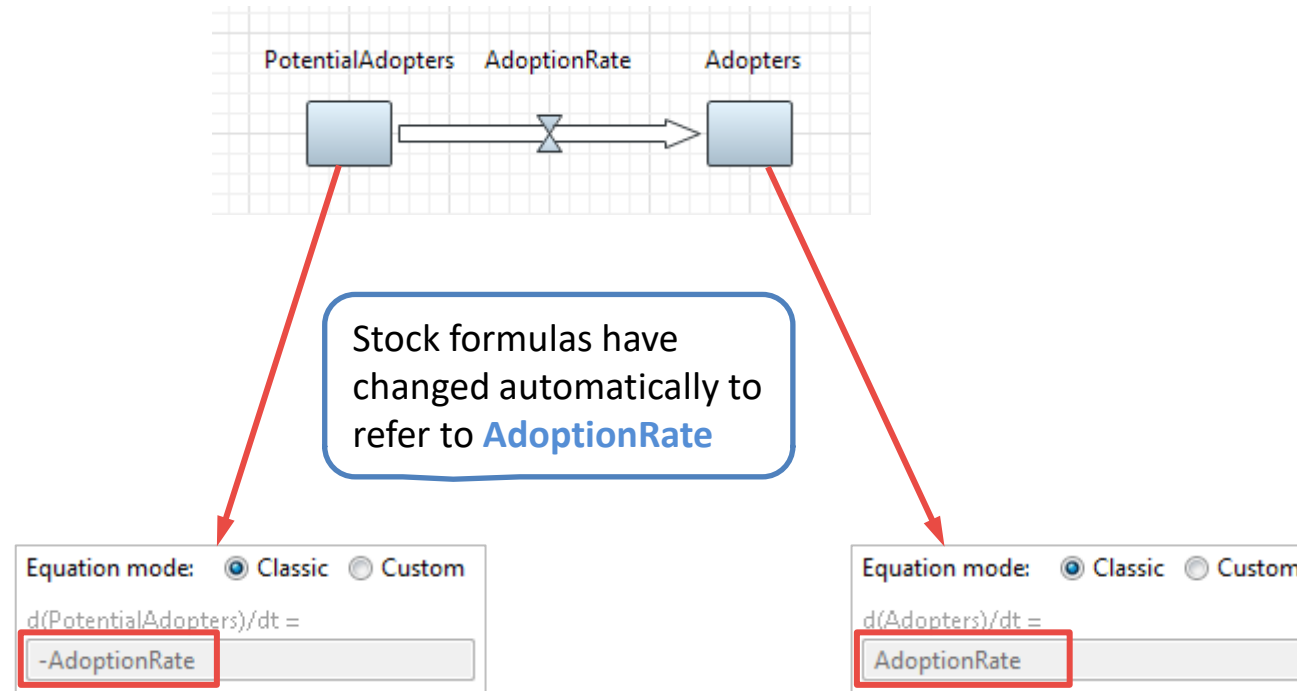
3 Click the stock where the flow flows in



The flow appears. If it is drawn correctly, its end points should be indicated with little green circles on selection.

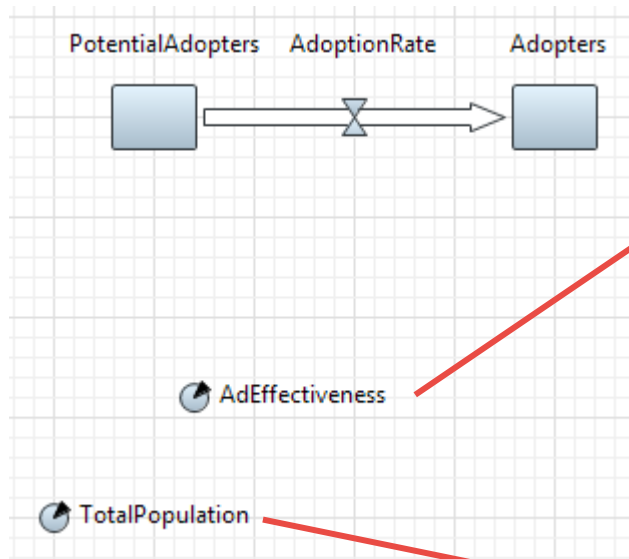
Bass Diffusion. Phase 1. Step 3

- Name the flow **AdoptionRate**
- Check the formulas of the stocks



Bass Diffusion. Phase 1. Step 4

- Create **TotalPopulation** and **AdEffectiveness** parameters



Properties X

AdEffectiveness - Parameter

Name:

☒ Show name ☐ Ignore

Visible: ☒ yes

Type:

Default value:

☐ System dynamics array

Properties X

TotalPopulation - Parameter

Name:

☒ Show name ☐ Ignore

Visible: ☒ yes

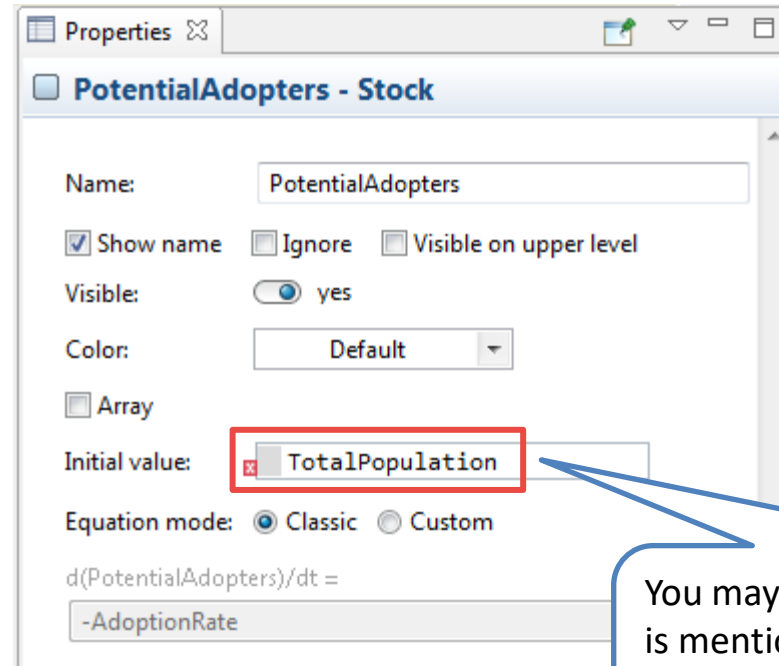
Type:

Default value:

☐ System dynamics array

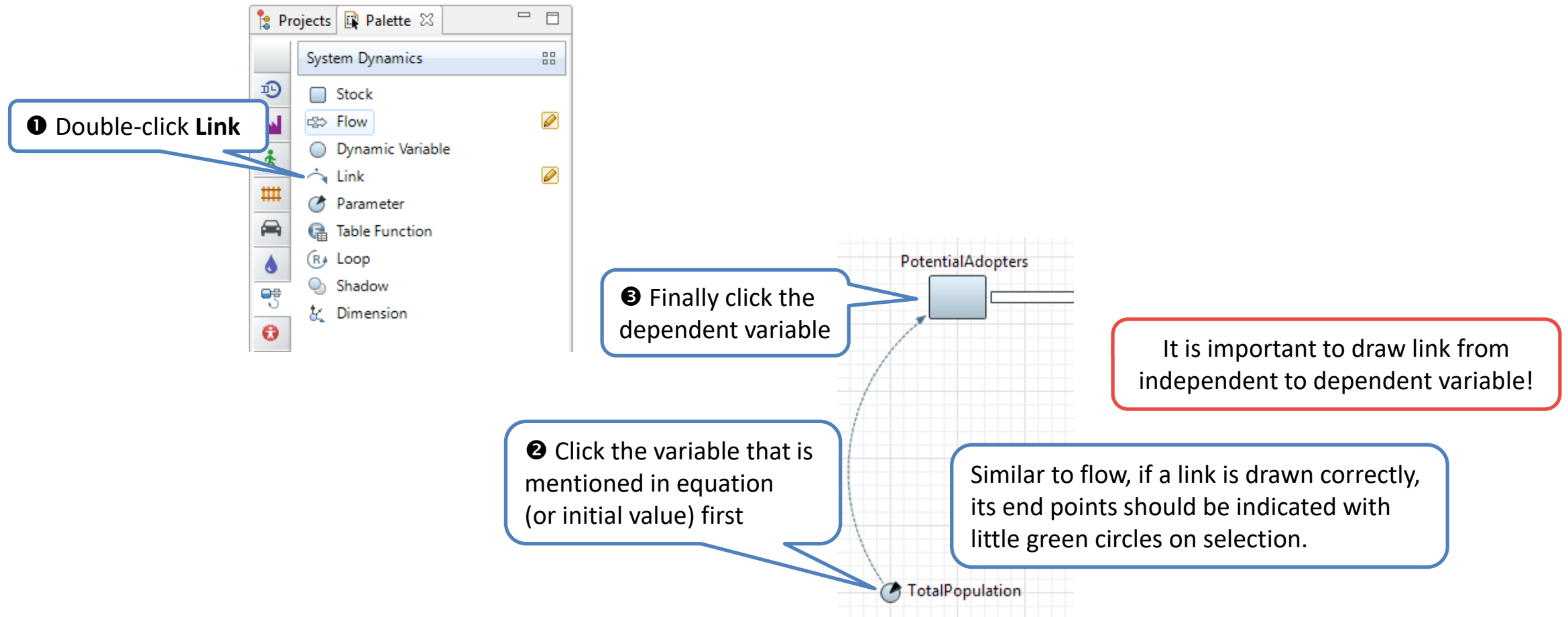
Bass Diffusion. Phase 1. Step 5

- Specify initial value for PotentialAdopters



Bass Diffusion. Phase 1. Step 6

- Draw a link defining causal dependency between **TotalPopulation** and **PotentialAdopters**



Bass Diffusion. Phase 1. Step 7

- Add **Dynamic Variable** *AdoptionFromAd*

AdoptionFromAd - Dynamic Variable

Name:

☒ Show name ☐ Ignore ☐ Visible on upper level

Visible: ☒ yes

Color:

☐ Array ☐ Dependent ☐ Constant

AdoptionFromAd=

☒ PotentialAdopters*AdEffectiveness

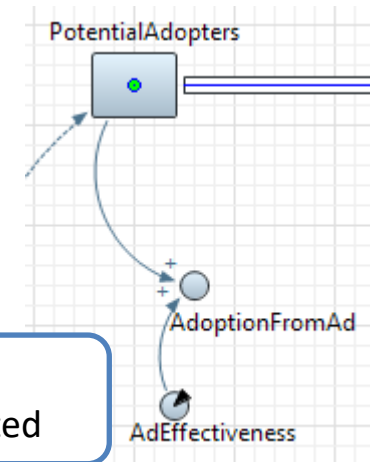
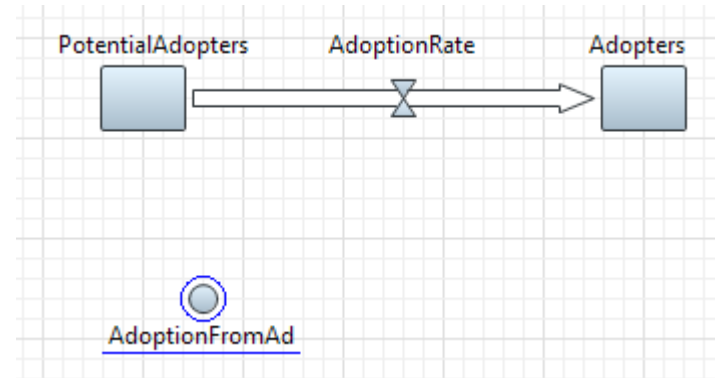
Create a link from AdEffectiveness

Create a link from PotentialAdopters

② Left-click the big error indicator

① Specify the formula and click inside the field

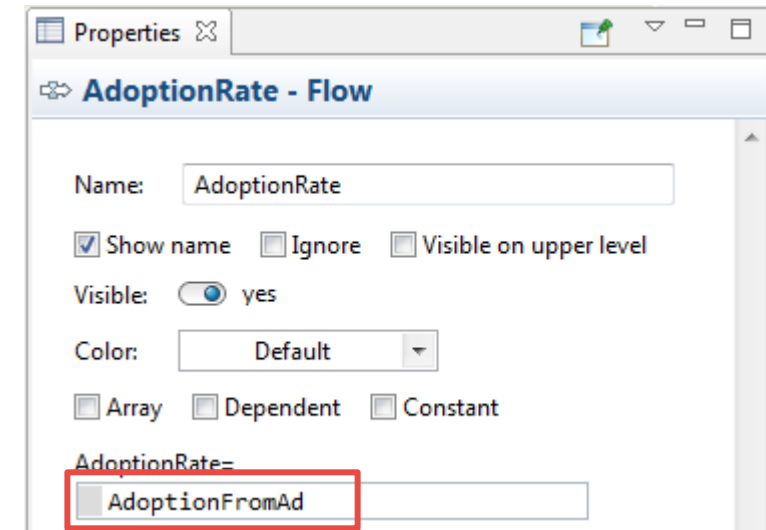
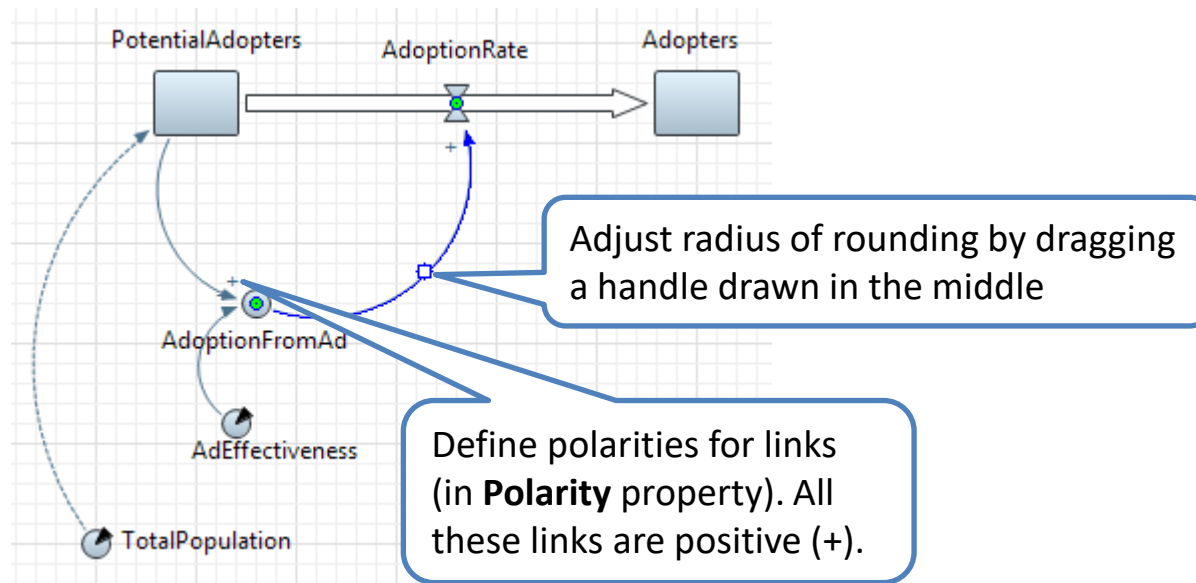
③ Create missing links by choosing corresponding menu items



You will see links automatically created

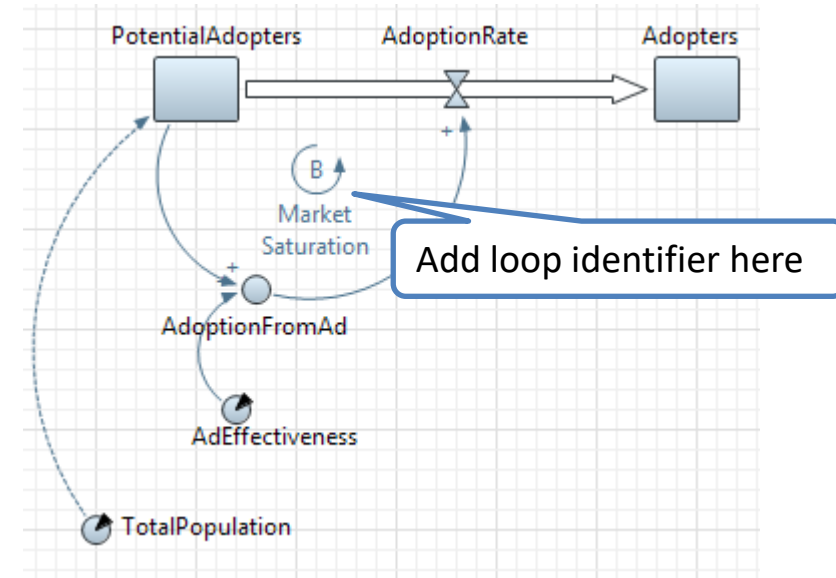
Bass Diffusion. Phase 1. Step 8

- Draw a link going from **AdoptionFromAd** to **AdoptionRate**
- Specify a formula for **AdoptionRate**



Bass Diffusion. Phase 1. Step 9

- Add **Loop** to indicate balancing causal loop caused by market saturation
- Set the loop direction, type and specify a brief description



Properties X

loop - Loop

Direction: ☐ Clockwise ☒ Counterclockwise

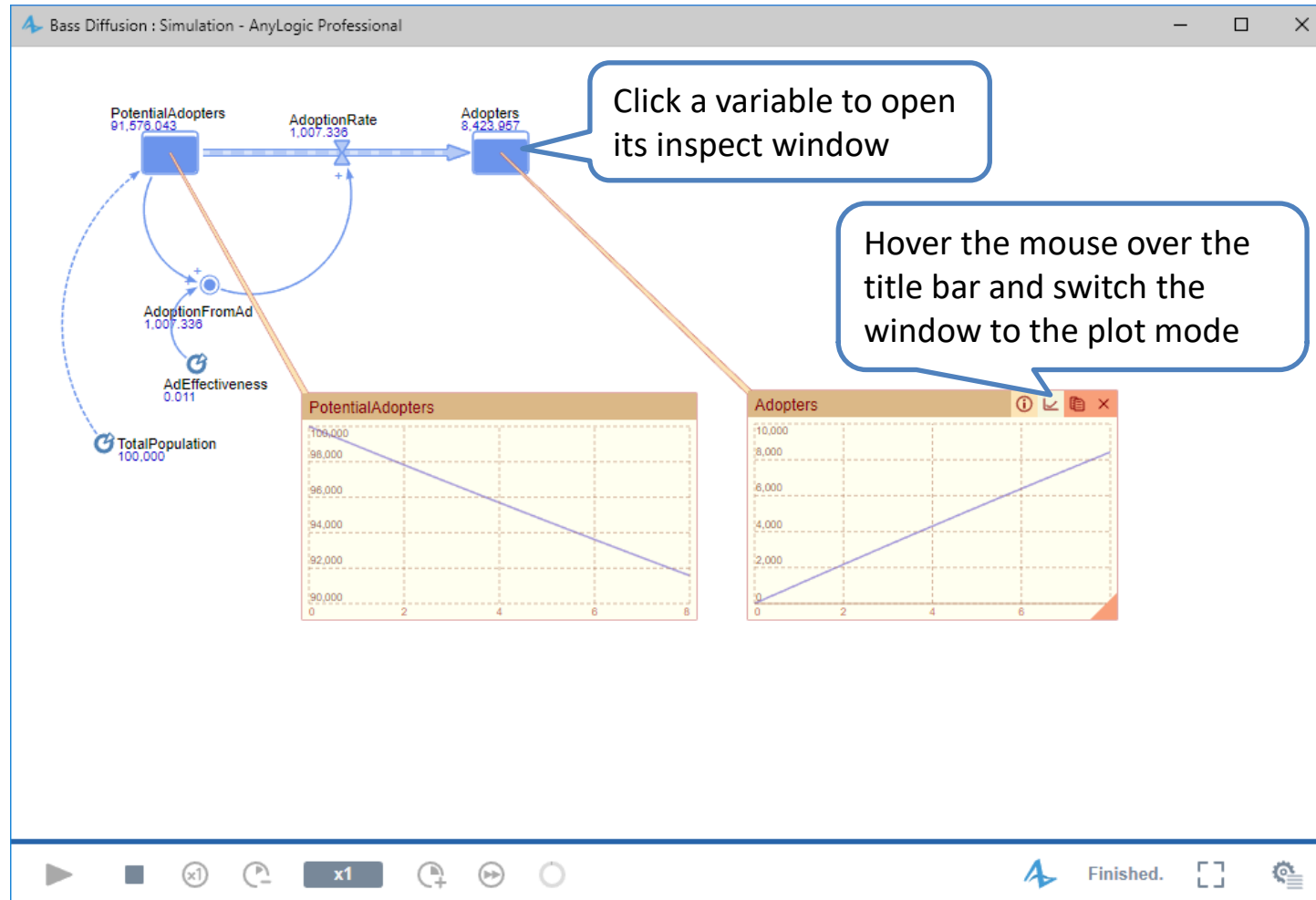
Type: ☒ B ☐ R ☐

Color:

Text:

Bass Diffusion. Phase 1. Step 10

- Run the model and observe the dynamics using inspect windows



Bass Diffusion. Phase 2

- Now add the effect of word of mouth on product diffusion
- Assume everybody contacts everybody else
- The number of person's contacts per time unit is **Contact Rate**
- If an adopter contacts a potential adopter, the last one will adopt with probability **Adoption Fraction**

- The corresponding part of the **Adoption Rate** will be:

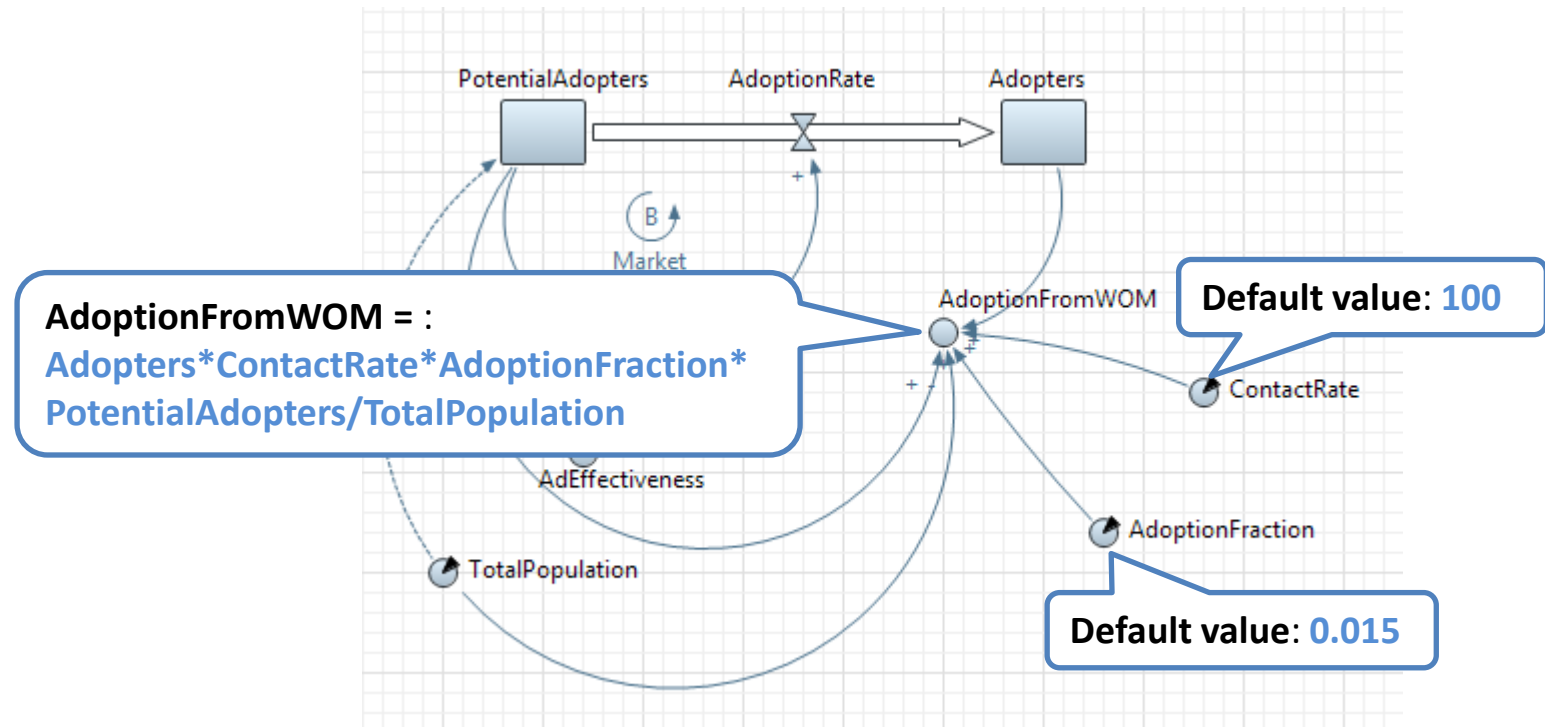
Adoption From Word of Mouth =
Adopters *
Contact Rate *
Adoption Fraction *
Potential Adopters / TotalPopulation

— Why?

- Parameter values:
 - *Contact Rate = 100*
 - *Adoption Fraction = 0.015*

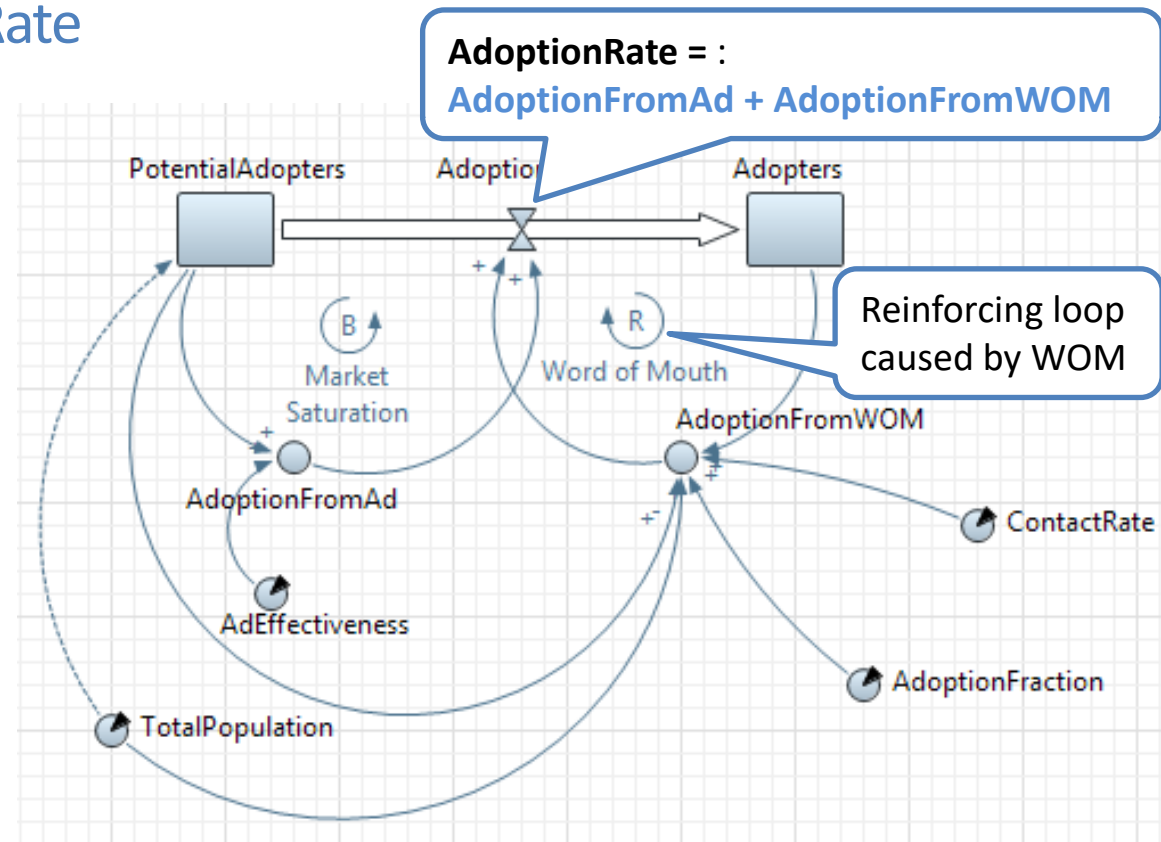
Bass Diffusion. Phase 2. Step 1

- Add **ContactRate** and **AdoptionFraction** parameters
- Add **AdoptionFromWOM** dynamic variable
- Draw dependency links

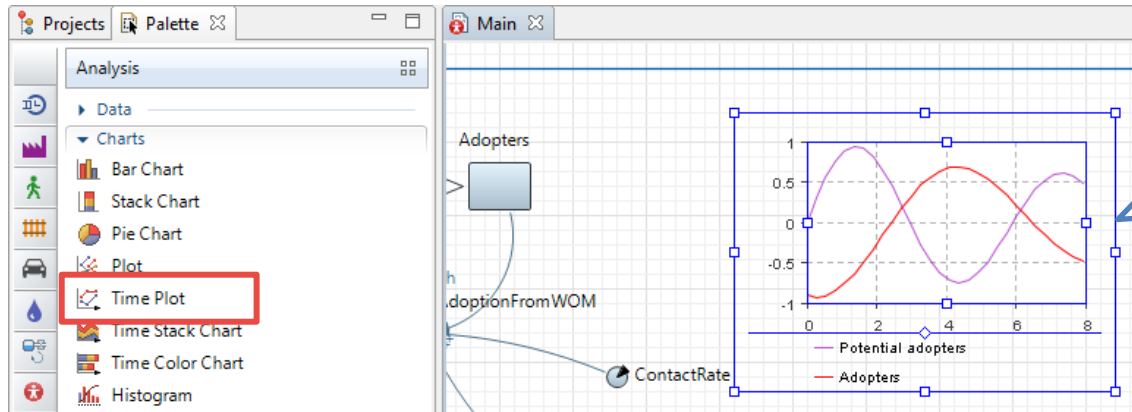


Bass Diffusion. Phase 2. Step 2

- Draw a link going from **AdoptionFromWOM** to **AdoptionRate**
- Modify the formula of **AdoptionRate**
- Add one more loop identifier



Bass Diffusion. Phase 2. Step 3



❶ Add **Time Plot** from **Analysis** palette to display how populations of potential adopters and adopters change in time

❷ Set **PotentialAdopters** and **Adopters** variables to be displayed by the plot

❸ Adjust the chart properties

Data

☒ Value ☐ Data set

Title: Potential adopters

Value: PotentialAdopters

Point style: [dropdown]

Line width: [dropdown] 1 pt

Color: mediumOrchid

☒ Value ☐ Data set

Title: Adopters

Value: Adopters

Point style: [dropdown]

Line width: [dropdown] 1 pt

Color: red

Data update

☒ Update data automatically
☐ Do not update data automatically

☒ Use model time ☐ Use calendar dates

First update time: [dropdown] 0 years

Update date: 31/10/2014 08:00:00

Recurrence time: [dropdown] 0.1 years

Display up to 100 latest samples (applies to "Value" data)

Scale

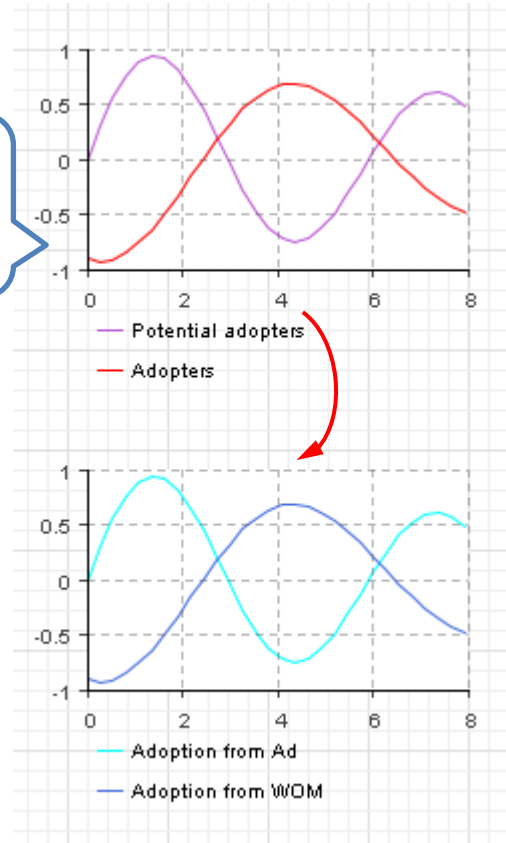
Time window: 8 model time units

Vertical scale: ☒ Auto ☐ Fixed

From: 0 To: 1

Bass Diffusion. Phase 2. Step 4

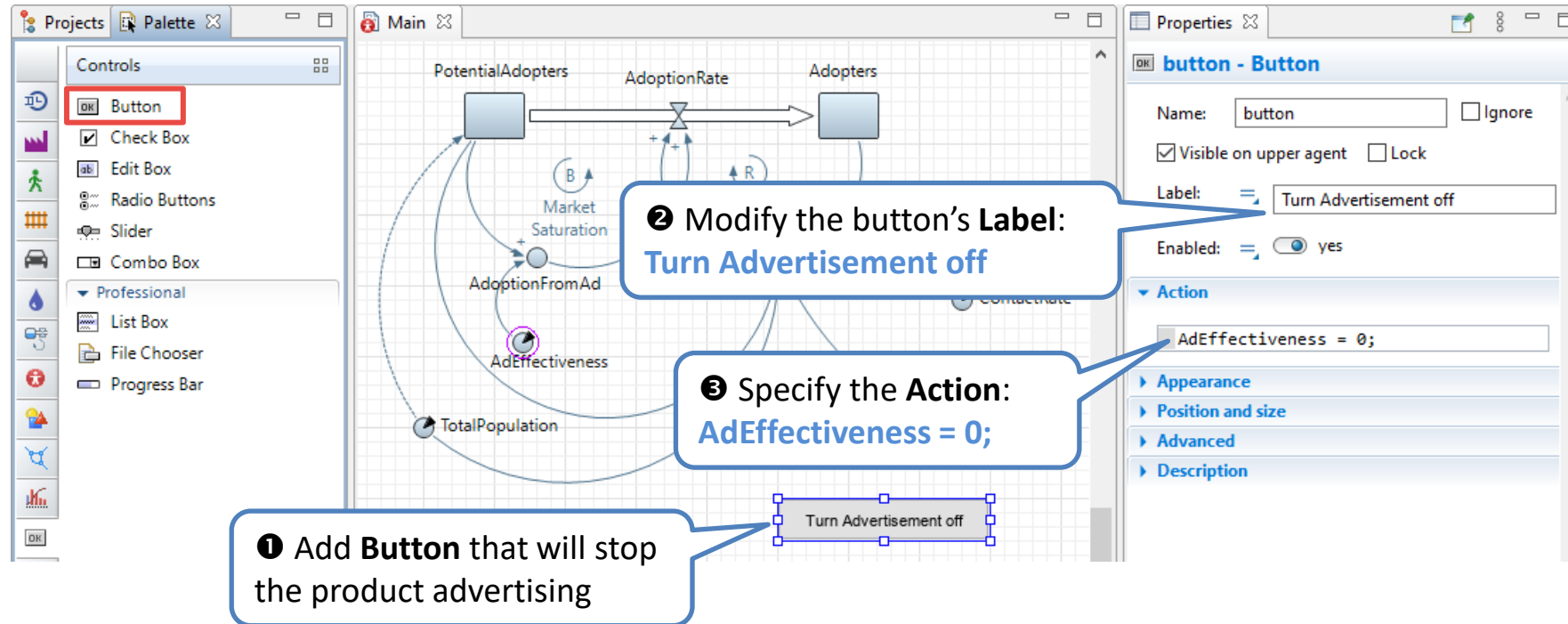
❶ Ctrl+drag (Mac OS: Cmd+drag) the plot to make a copy



The screenshot shows the 'Properties' window for 'plot1 - Time Plot'. The 'Data' section is expanded, showing two data series. The first series is 'Adoption from Ad' with a cyan color. The second series is 'Adoption from WOM' with a royalBlue color. A red box highlights the 'Data' section, and a blue callout points to the 'Adoption from WOM' series.

❷ Add **AdoptionFromAd** and **AdoptionFromWOM** variables onto the plot

Bass Diffusion. Phase 2. Step 5



Bass Diffusion. Phase 2. Step 6

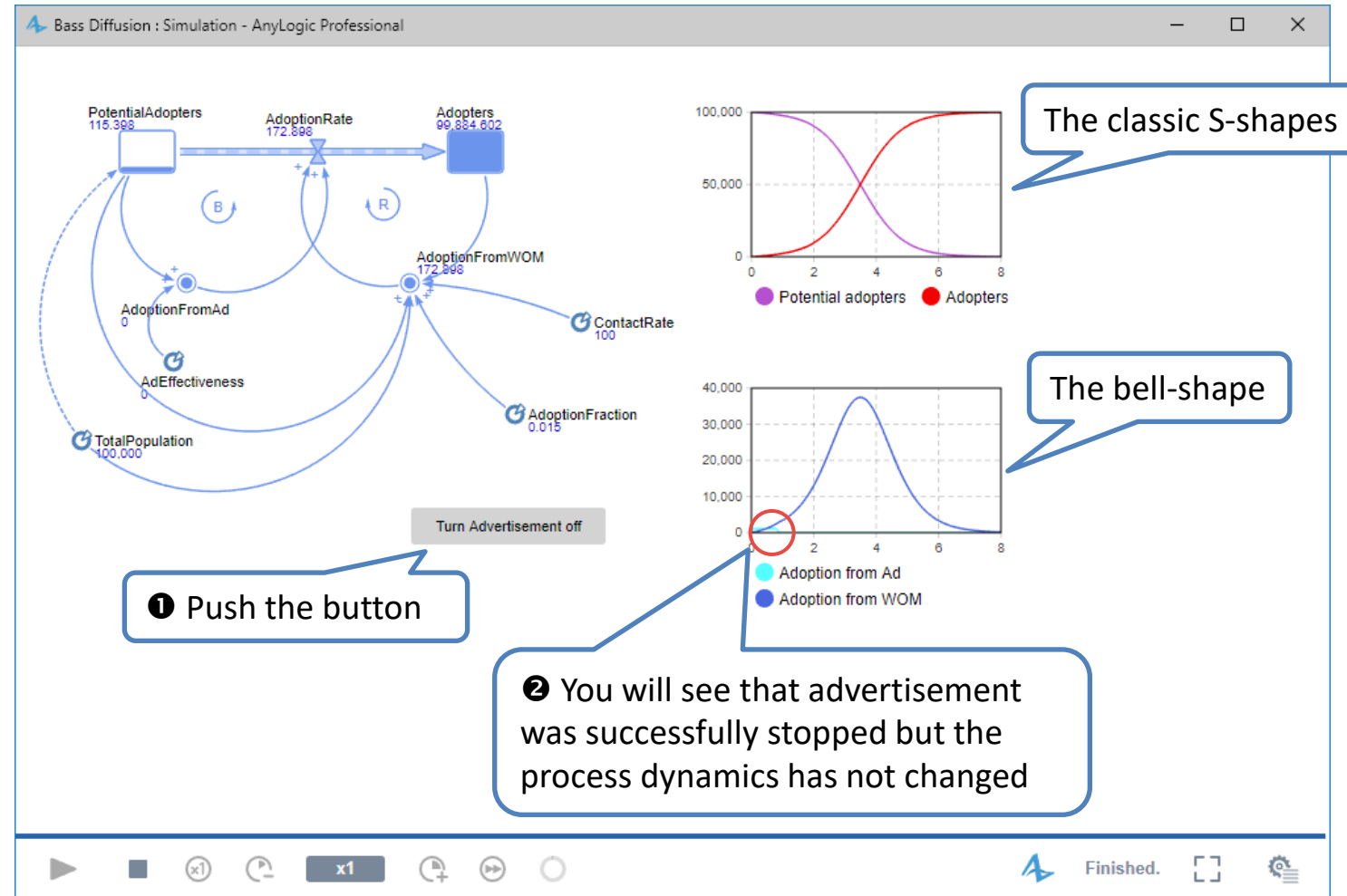
- Set the model to stop after 8 time units

The screenshot displays the AnyLogic software interface with three main panels: Projects, Main, and Properties.

- Projects Panel:** Shows a tree view with 'Bass Diffusion' expanded, containing 'Main' and 'Simulation: Main'. A callout bubble with a blue border and a black circle containing the number 1 points to the 'Simulation: Main' item, with the text '1 Click Simulation item'.
- Main Panel:** Shows a grid with the text 'Bass Diffusion'. A callout bubble with a blue border and a black circle containing the number 2 points to the 'Simulation' tab, with the text '2 Click Model Time tab'.
- Properties Panel:** Shows the 'Simulation - Simulation Experiment' configuration. The 'Model time' section is expanded. The 'Stop:' dropdown is set to 'Stop at specified time' and is highlighted with a red rectangle. The 'Start time:' is set to 0 and the 'Stop time:' is set to 8. A callout bubble with a blue border and a black circle containing the number 3 points to the 'Stop time:' field, with the text '3 Set the experiment to stop at specified time: 8'.

Bass Diffusion. Phase 2. Step 7

- Run the model and observe the dynamics using plots
- Stop advertisement and watch how it affected the adoption process



Bass Diffusion. Phase 2. Questions

1. In the 80s, he was the System Dynamics enthusiast:
 - (a) David Hockney, Painter
 - (b) Milos Zeman, former Czech Republic President
 - (c) Anthony Lake, Executive Director of the UNICEF
2. Please explain why one loop is balancing and another – reinforcing.

Parameter Variation Experiment in AnyLogic Cloud

- Now we want to study how adoption process depends on the adopters' contact rate
- We will use [parameter variation experiment](#)
- This experiment performs multiple model runs varying one or more parameters of the model

AnyLogic Cloud. Step 1

- Set up the model prior to uploading it to the AnyLogic Cloud

The screenshot displays the AnyLogic software interface. On the left, the 'Projects' pane shows a tree view for a project named '>Bass Diffusion*'. Under this project, there are three items: 'Main', 'Simulation: Main', and 'Run Configuration: Main'. A blue callout box with the number '1' points to the 'Run Configuration: Main' item, containing the text: '1 Double-click the Run Configuration item'. The main workspace is divided into two panes. The top pane, titled 'Main', shows a tree view of the model's components. Under the 'Inputs' folder, four parameters are listed: 'TotalPopulation: 100000', 'ContactRate: 100', 'AdEffectiveness: 0.011', and 'AdoptionFraction: 0.015'. A red box highlights these four parameters, and a red arrow points from this box to the 'Inputs' panel on the right. The bottom pane, titled 'Run Configuration', shows the 'Inputs' and 'Outputs' panels. The 'Inputs' panel has a dashed border and contains the text 'Drag and drop inputs here'. The 'Outputs' panel also has a dashed border and contains the text 'Drag and drop outputs here'. A blue callout box with the number '2' points to the 'Inputs' panel, containing the text: '2 Add input parameters to the model's workbench by dragging them into the panel on the right'.

1 Double-click the Run Configuration item

2 Add input parameters to the model's workbench by dragging them into the panel on the right

AnyLogic Cloud. Step 2

The image shows two windows from the AnyLogic software: 'Run Configuration' and 'Properties'.

Run Configuration Window:

- Inputs:** A list of model inputs including 'TotalPopu', 'ContactR', 'AdEffectiv', and 'Adoption'.
- Outputs:** A list of model outputs including 'plot: Potential ad...', and 'plot1: Adoption from A...'. The 'plot1: Adoption from A...' item is highlighted with a red box.
- Outputs Section:** A section at the bottom of the window titled 'Outputs' with a bar chart icon. It contains two entries: 'Plot' and 'Plot 1', each with a corresponding plot icon and a close button.

Properties Window:

- Export model to AnyLogic Cloud:** A link at the top of the window.
- Top-level agent:** A dropdown menu set to 'Main'.
- Maximum:** A field for the maximum number of agents.
- Model time:** A section with a dropdown menu set to 'Stop at specified time'.
- Stop:** A dropdown menu set to 'Stop at specified time'.
- Start time:** A text field with the value '0'.
- Stop time:** A text field with the value '8'.

Callouts:

- ① Add the model's charts to the model's **Outputs** section
- ② Choose **Stop: Stop at specified time, Stop time: 8**
- ③ Click the **Export model to AnyLogic Cloud** label

AnyLogic Cloud. Step 3

Export model to AnyLogic Cloud

Upload your model to AnyLogic Cloud

anylogic cloud

Log in or [register new account](#).

E-mail: Password:

☐ Remember me

Log in / Sign up with: 

< Back Next > Finish Cancel

❶ Register at AnyLogic Cloud

Export model to AnyLogic Cloud

Upload your model to AnyLogic Cloud

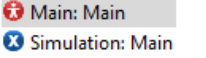
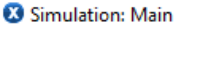
anylogic cloud

User: User [change...](#)

Upload as a new model or choose the existing one to upload a new version.

Model:

Model name:

Icon:  

☐ Include model source files
Depending on the sharing settings when uploading a model in AnyLogic Cloud these files may become public.

☐ Provide public access to this model

< Back **Next >** Finish Cancel

❷ Upload new model to the Cloud

AnyLogic Cloud. Step 4

AnyLogic Cloud

AnyLogic North America, LLC. [US] | <https://cloud.anylogic.com/#/model/3e3beba9-3962-4a65-980d-594cc514d570;mode=DAS...>

Bass Diffusion > 1 > Experiment

1 23.11.2017, 16:39:55 +

Experiment
23.11.2017 | Simulation

Inputs

Total population

Contact rate 100

Ad effectiveness 0.011

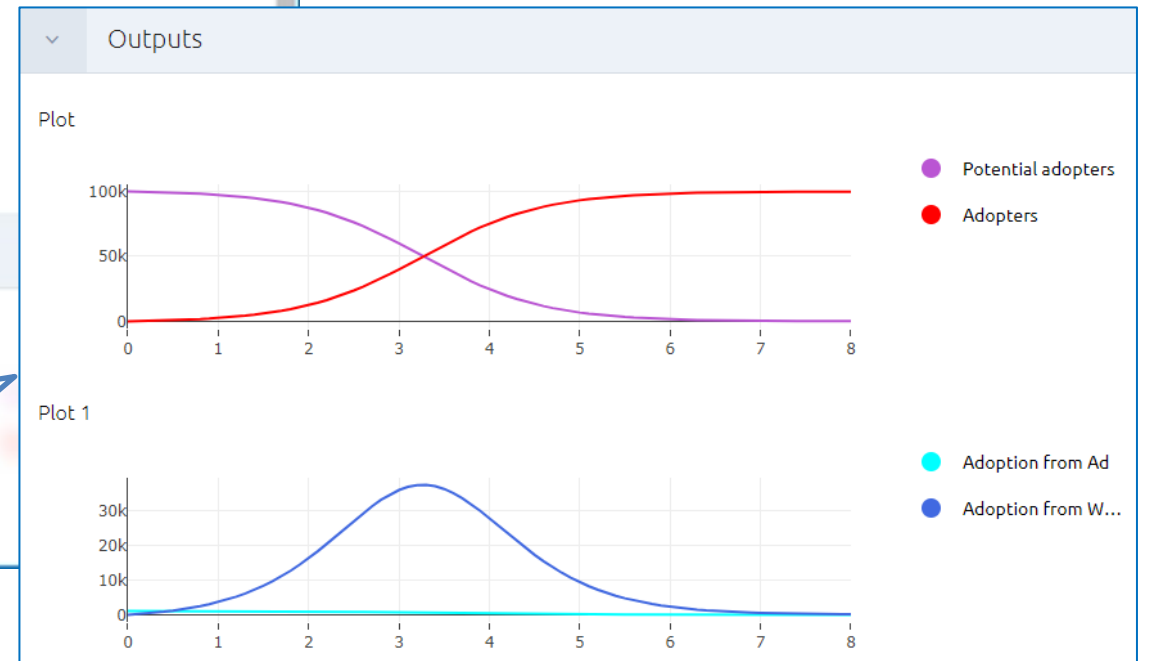
Adoption fraction 0.015

Outputs

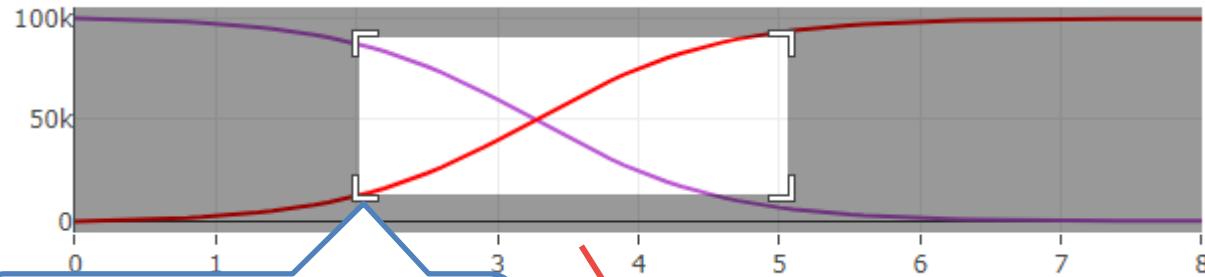
① Click here to open the simulation experiment page

② Click the **Run** button to run the simulation experiment

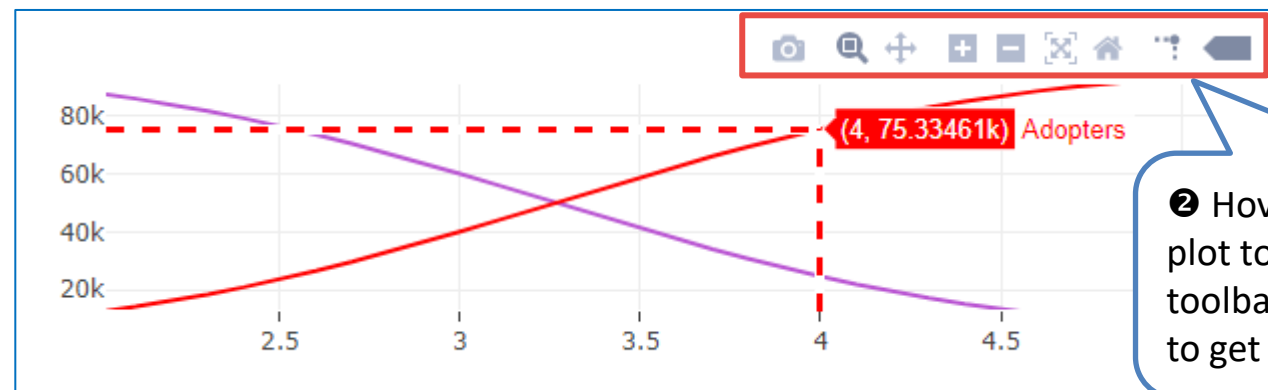
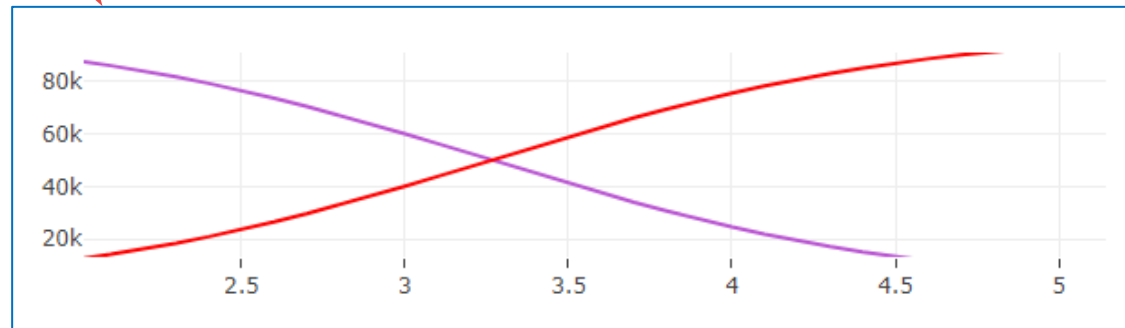
③ Observe the **Outputs** section



AnyLogic Cloud. Step 5

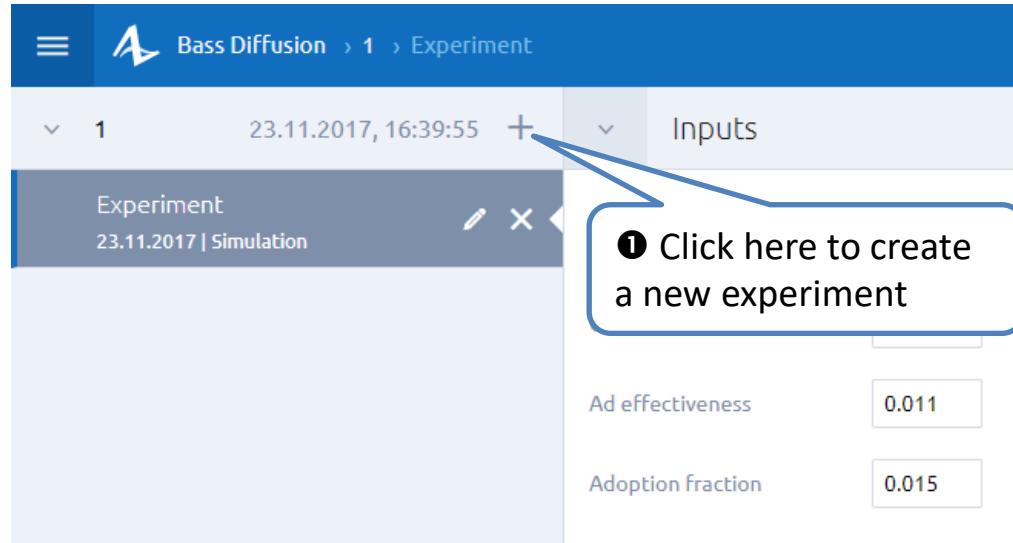


❶ To zoom in to a specific part of the chart, select this area by dragging the mouse



❷ Hover the mouse over the plot to see the chart settings toolbar. Play with the options to get more from the chart.

AnyLogic Cloud. Step 6



2 Select the **Variation** experiment type

The 'New experiment' dialog box is shown. It has a title bar with a close button (X). The main content area includes: 'Version 1' (text), 'Experiment name' (text) with a text input field containing 'ContactRateVariation', 'Experiment type' (text) with a dropdown menu set to 'Variation', and a description: 'Multiple iterations with one or more inputs varied within the specified range with the given step.' At the bottom right, there are two buttons: 'Cancel' (light blue) and 'Add' (dark blue).

AnyLogic Cloud. Step 7

1 One more experiment will appear in the panel

2 Click here to configure the experiment

4 Save the changes

3 Set the **Contact rate** parameter to be **Varied in range**

AnyLogic Cloud. Step 8

Inputs

Setup ranges, step

Total population 100000

Contact rate - step

Ad effectiveness 0.011

Adoption fraction 0.015

❶ Set up the min, max and step values for the **Contact rate** parameter

❷ Run the variation experiment



❸ Observe how adoption process depends on the contact rate

Questions

1. What will happen to the final number of Adopters If Contact Rate grows?
2. How many Adopters will there be in the end if Contact Rate equals zero:
 - (a) 0 adopters
 - (b) more than 1
 - (c) more than half of all people

Exercise. SEIR Model

- Model dynamics of contagion:
 - A **Susceptible** person, being infected, becomes **Exposed**, then after incubation time, **Infectious**, and then recovers
 - Both Exposed and Infectious people can pass the disease to Susceptible, but they have different infectivity (probability of passing disease during contact) and different contact rate
 - **Recovered** become immune to the disease
 - Parameters:
 - TotalPopulation**: 10000
 - ContactRateExposed**: 4, **ContactRateInfectious**: 1.25
 - InfectivityExposed**: 0.05, **InfectivityInfectious**: 0.06
 - AverageIncubationTime**: 10 days
 - AverageIllnessDuration**: 15 days
 - Initially: one person is Exposed, all others – Susceptible